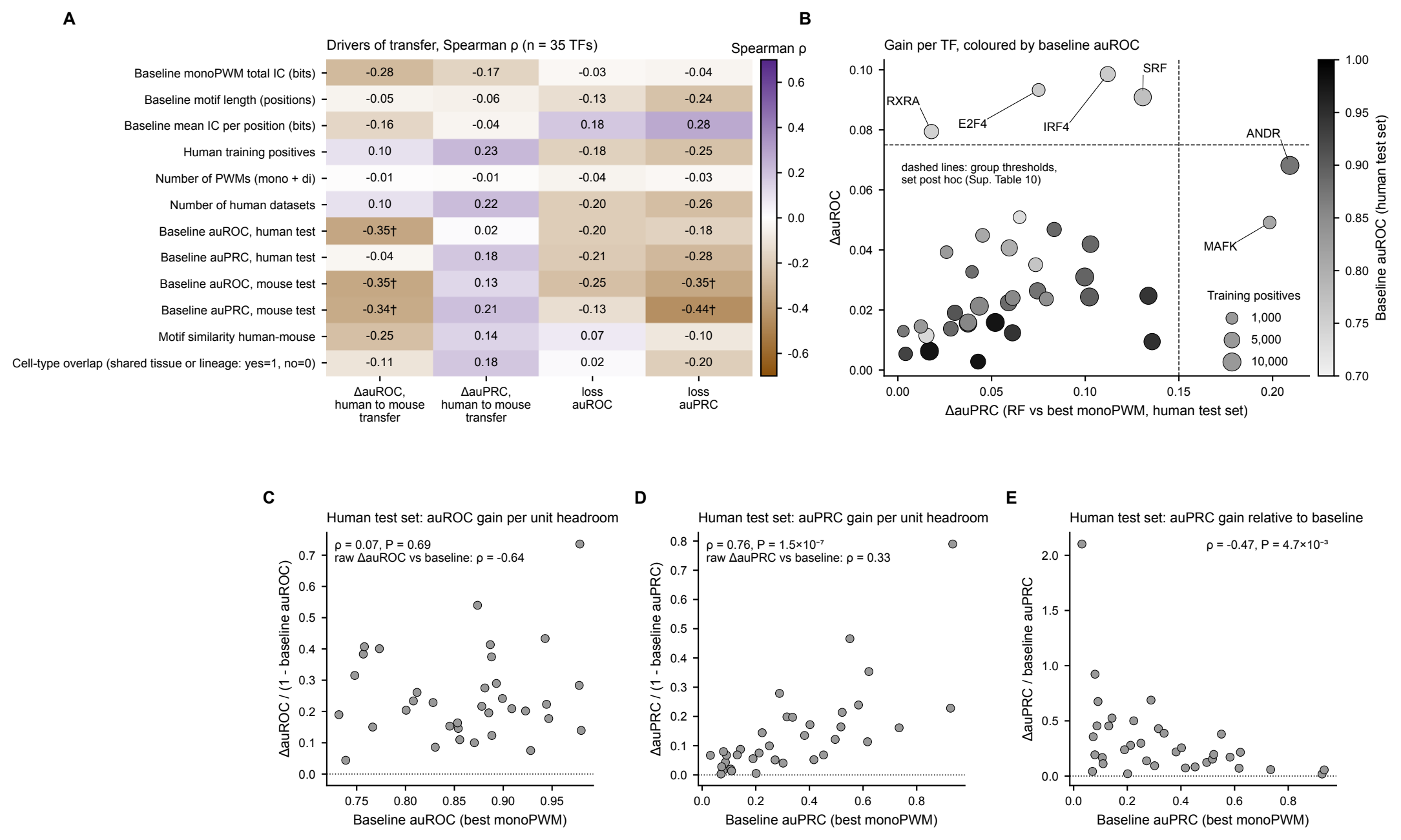

A
Drivers of transfer, Spearman ρ (n = 35 TFs)
Spearman ρ
Baseline monoPWM total IC (bits)
Baseline motif length (positions)
Baseline mean IC per position (bits)
Human training positives
Number of PWMs (mono + di)
Number of human datasets
Baseline auROC, human test
Baseline auPRC, human test
Baseline auROC, mouse test
Baseline auPRC, mouse test
Motif similarity human-mouse
Cell-type overlap (shared tissue or lineage: yes=1, no=0)
-0.28 -0.17 -0.03 -0.04
-0.05 -0.06 -0.13 -0.24
-0.16 -0.04 0.18 0.28
0.10 0.23 -0.18 -0.25
-0.01 -0.01 -0.04 -0.03
0.10 0.22 -0.20 -0.26
-0.35† 0.02 -0.20 -0.18
-0.04 0.18 -0.21 -0.28
-0.35† 0.13 -0.25 -0.35†
-0.34† 0.21 -0.13 -0.44†
-0.25 0.14 0.07 -0.10
-0.11 0.18 0.02 -0.20
ΔauROC, human to mouse transfer
ΔauPRC, human to mouse transfer
loss auROC
loss auPRC
B
Gain per TF, coloured by baseline auROC
RXRA
E2F4
IRF4
SRF
ANDR
MAFK
dashed lines: group thresholds, set post hoc (Sup. Table 10)
Training positives
1,000
5,000
10,000
ΔauROC
ΔauPRC (RF vs best monoPWM, human test set)
Baseline auROC (human test set)
C
Human test set: auROC gain per unit headroom
ρ = 0.07, P = 0.69
raw ΔauROC vs baseline: ρ = -0.64
ΔauROC / (1 - baseline auROC)
Baseline auROC (best monoPWM)
D
Human test set: auPRC gain per unit headroom
ρ = 0.76, P = 1.5×10⁻⁷
raw ΔauPRC vs baseline: ρ = 0.33
ΔauPRC / (1 - baseline auPRC)
Baseline auPRC (best monoPWM)
E
Human test set: auPRC gain relative to baseline
ρ = -0.47, P = 4.7×10⁻³
ΔauPRC / baseline auPRC
Baseline auPRC (best monoPWM)